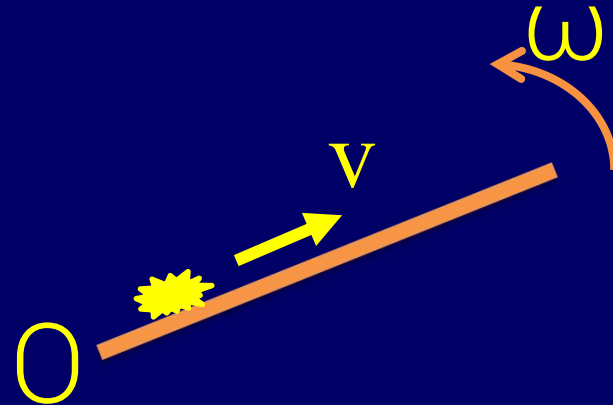


Quiz 1: A rod rotates with an angular speed ω around one of the rod end O on the horizontal plane. An insect moves away from the fixed end O with the constant speed v along the rod at $t=0$. Calculate the acceleration speed of the insect relative to the ground.



Answer:

$$\mathbf{v} = \omega r \mathbf{e}_\theta + v \mathbf{e}_r$$

$$\mathbf{a} = \frac{d\mathbf{v}}{dt} = \omega \frac{dr}{dt} \mathbf{e}_\theta + \omega r \frac{d\mathbf{e}_\theta}{dt} + v \frac{d\mathbf{e}_r}{dt}$$

$$\begin{aligned} \mathbf{a} &= \omega v \mathbf{e}_\theta - \omega^2 r \mathbf{e}_r + v \omega \mathbf{e}_\theta \\ &= 2\omega v \mathbf{e}_\theta - \omega^2 r \mathbf{e}_r \end{aligned}$$